

Scalability & Future Plan

Date	01 October 2026
Team ID	SWTID-2026-9629
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

How PocketSmart AI can scale to more users, more data and more features, and the roadmap beyond the current version.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Recommendations are search links and AI price estimates, not live platform data.	Prices and availability may differ from real listings.	High
2	Every plan waits for a synchronous Gemini call.	Slower or failed plans under heavy use or quota limits.	High
3	Runs on the Flask development server with a single SQLite file.	Limited concurrent users and storage growth.	Medium
4	No CSRF protection, rate limiting or HTTPS.	Not ready for public deployment.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask development server	Run under Gunicorn with several workers behind a reverse proxy; containerise with Docker.
2	Data Storage	SQLite file (pocketsmart.db)	Switch DATABASE_URL to PostgreSQL and add database migrations.
3	Performance	Each request calls Gemini and waits	Cache frequent plans and run long AI requests as background jobs.
4	Security	Session login and .env keys	Add CSRF protection, rate limiting, stronger password rules, HTTPS and secure cookies.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Use official or affiliate APIs for live product prices and availability	1-2 months	Accurate, up-to-date budget-fit recommendations
Phase 3	Add more planners (travel, wedding, electronics, groceries) and regional-language support	3-4 months	Wider audience and more use cases
Phase 4	Mobile app, plan comparison and expense tracking from saved history	6+ months	Higher engagement and long-term budgeting support